

Algebraic structures

1. OPERATION [a function $\ell: A \times A \rightarrow A$.

$(A, \cdot)$ generic notation for an arbitrary operation.

$+, \cdot$ on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C} \rightarrow$ operations $\checkmark$

$-$ on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
 $\times \checkmark \checkmark \checkmark \checkmark$

$:$ on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C} \rightarrow$ because of 0. (we cannot divide by 0)
 $\times \times \times \times \times$

1.1. OPERATION LAWS

$\rightarrow$ associativity law

$\rightarrow$ commutativity law

$\rightarrow$ identity law (e.m. = id.el.) $\rightarrow$ unique id. elem.

$\rightarrow$ inverse law (el.sim. = inv.el.)

{ associative

id. elem

$\Rightarrow$ unique inv. el.

} inv. elem.

2. STABLE SUBSET $\ell: A \times A \rightarrow A$

$$B \subseteq A$$

B stable subset of A if $(\forall) x, y \in B, \ell(x, y) \in B$.

the operation induced by $\ell: \ell': B \times B \rightarrow B, \ell'(x, y) = \ell(x, y)$

① Is $2\mathbb{Z} = \{2k \mid k \in \mathbb{Z}\}$ stable in $(\mathbb{Z}, +)$? $\checkmark$ ($2+2=4$)

$2\mathbb{Z} + 1 = \{2k+1 \mid k \in \mathbb{Z}\}$ stable in $(\mathbb{Z}, +)$? $\times$ ($1+5=6$)

② Is $[0, 1]$ stable in $(\mathbb{R}, \cdot)$? $\checkmark$

$[-1, 0]$ stable in $(\mathbb{R}, \cdot)$? $\times$ ($-1 \cdot (-1) = 1 \notin [-1, 0]$)

associativity + commutativity still true in the stable subset. $(\forall)$

id elem + inv. elem not always true in the stable subset. $(\exists)$

3. GROUPS

3.1. SEMIGROUP

→ assoc.

3.2. MONOID

→ assoc.

→ id. el.

3.3. GROUP

→ assoc.

→ id. el.

→ (†) el. has an inv. el.

$1 \rightarrow$ notation for id. el.
 $x^{-1} \rightarrow$ notation for inv. el.
 $\left. \begin{array}{l} \phantom{x^{-1} \rightarrow \text{notation for inv. el.}} \\ \phantom{x^{-1} \rightarrow \text{notation for inv. el.}} \end{array} \right\} (G, \cdot)$

$0 \rightarrow$ notation for id. el.
 $-x \rightarrow$ notation for inv. el.
 $\left. \begin{array}{l} \phantom{-x \rightarrow \text{notation for inv. el.}} \\ \phantom{-x \rightarrow \text{notation for inv. el.}} \end{array} \right\} (G, +)$

3.4. POWERS IN GROUPS

$(G, \cdot)$ semigroup. $x^m = \underbrace{x \cdot x \cdot \dots \cdot x}_{m \text{ times}}$

$(G, \cdot)$ monoid: $x^0 = 1$.

$(G, \cdot)$ group: $x^{-m} = (x^{-1})^m$

$(G, +)$ semigroup. $x^m = \underbrace{x + x + \dots + x}_{m \text{ times}} = mx$

3.5. PROPERTIES

$(G, \cdot)$ group.

1) $x^m \cdot x^n = x^{m+n}$

2) $(x^m)^n = x^{m \cdot n}$

Cancellation laws:

$$1) a^{-1} \cdot a \cdot x = a \cdot y \Rightarrow x = y$$

$$x \cdot a = y \cdot a \cdot a^{-1} \Rightarrow x = y$$

$$2) (x^{-1})^{-1} = x$$

$$3) (x \cdot y)^{-1} = y^{-1} \cdot x^{-1}$$

Finite group $\Rightarrow$ "tabla legii" $\Rightarrow$ no repetitions on each row/column.

3.6. EXAMPLES

a) $(\mathbb{Z}, -)$ not assoc.

$$(1-2)-3 = -1-3 = -4$$

$$1-(2-3) = 1-(-1) = 2$$

b) $X \neq \emptyset$

a word on X of length $m \rightarrow$ a string of m elements from X .

the word of length 0 = the void word = e .

$(X^*, \cdot)$ monoid with id. $e = e \rightarrow$ the FREE MONOID

c) division is not an operation

d) raising to the power is not an operation

$$(a^b)^c \neq a^{b^c}$$

e) $(\mathbb{Z}_m, +)$, $m \in \mathbb{N}_2$ abelian group.

b) GROUP OF RESIDUE CLASSES modulo m .

$$\hat{x} + \hat{y} = \widehat{x+y}$$

$$x \% m + y \% m = (x+y) \% m.$$

f) matrices

g) permutations

3.7. KLEIN'S GROUP

$$K = \{e, a, b, c\}$$

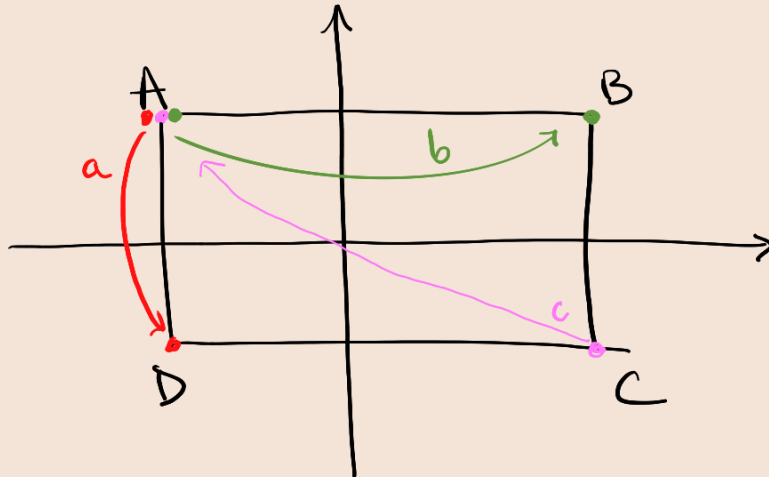
$$R = \{e, a, b, c\}$$

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

$\Rightarrow e$ id. elem.

$\Rightarrow a = a^{-1}; b = b^{-1}; c = c^{-1}$.

.....



e ? identical transf.
where on drawing?

4. RINGS

$(R, +, \cdot)$ ring if

1) $(R, +)$ abelian group

2) $(R, \cdot)$ semigroup $\rightarrow$ monoid?

3) distributive laws

$(R, +, \cdot)$ UNITARY RING

$1 \neq 0$ (the ones we talked at school)

1) $(R, +)$ abelian group

2) $(R, \cdot)$ monoid

3) distrib. laws

$(R, +, \cdot)$ DIVISION RING
SKEW FIELD
(corp)

1) $(R, +)$ abelian group $\rightarrow |R| \geq 2$

2) $(R^*, \cdot)$ group $\rightarrow (\forall) x \in R^*, \exists x^{-1} \in R^*$

3) distrib. laws

$(R, +, \cdot)$ FIELD

(corp commutativ)

1) $(R, +)$ abelian group

2) $(R^*, \cdot)$ abelian group

3) distrib. laws

! $(R, +, \cdot)$ com. if " $\cdot$ " is comm.

! $(R, +, \cdot)$ ring

id.el: 0 id.el: 1.

$$! R^* = R \setminus \{0\}$$

4.1. POWERS IN RINGS

$$\begin{aligned} m \cdot x &= \underbrace{x + \dots + x}_{m \text{ times}} \\ \text{digit} \swarrow \quad \searrow \text{id. elem. } (R, +) \\ 0 \cdot x &= 0 \end{aligned}$$

$$(-m) \cdot x = -m \cdot x$$

$$x^m = \underbrace{x \cdot \dots \cdot x}_{m \text{ times}}$$

$$x^0 = 1 \quad] \text{ unitary ring (}$$

$$x^{-m} = (x^{-1})^m \quad] \text{ division ring (corp)}$$

4.2. PROPERTIES IN RINGS

$$1) x(y-z) = x \cdot y - x \cdot z \dots$$

$$2) x \cdot 0 = 0 \cdot x = 0$$

$$3) x \cdot (-y) = (-x) \cdot y = -x \cdot y$$

Examples

a) $(\mathbb{Z}, +, \cdot)$ unitary ring;
not a field

b) $(\mathbb{N}, +, \cdot)$ not a ring; $(\mathbb{N}, +)$ is not a group.

c) $(\mathbb{Z}_m, +, \cdot)$ field $\Leftrightarrow m$ prime

d) polynomials

e) matrix

f) $(R, +, \cdot)$ ring, $M \neq \emptyset$

$$R^M = \{ f \mid f: M \rightarrow R \}$$

$$(f+g)(x) = f(x) + g(x)$$

$$(\forall) f, g \in R, f+g: M \rightarrow R, (f+g)(x) = f(x) + g(x)$$

$$f \cdot g: M \rightarrow R, (f \cdot g)(x) = f(x) \cdot g(x)$$

$(R^M, +, \cdot)$ ring of functions.

g) Boolean ring

5. SUBGROUPS

$(G, \cdot)$ group. $H \leq G$

H is a subgroup if: 1) H is a stable subset of $(G, \cdot)$

2) $(H, \cdot)$ is a group $\rightarrow$ ass.
 $\rightarrow$ id. elem.
 $\rightarrow$ inv. el.

$H \leq G$ notation.

More efficient (Andrea's style)

$$H \leq G \Leftrightarrow \begin{cases} H \neq \emptyset \quad (1 \in H) \\ (\forall) x, y \in H, x \cdot y \in H \\ (\forall) x \in H, x^{-1} \in H. \end{cases} \quad (\Leftrightarrow) \begin{cases} H \neq \emptyset \quad (1 \in H) \\ (\forall) x, y \in H, x \cdot y^{-1} \in H. \end{cases}$$

$$\underline{(G, +)}$$
 same thing but $\begin{cases} H \neq \emptyset \quad (1 \in H) \\ (\forall) x, y \in H, x + y \in H \\ (\forall) x \in H, -x \in H \end{cases} \quad (\Leftrightarrow) \begin{cases} H \neq \emptyset \quad (1 \in H) \\ (\forall) x, y \in H, x - y \in H. \end{cases}$

Examples:

1) $(G, \cdot)$ has trivial subgroups: $\{1\}$ and G .

2) $\mathbb{Z}$ subgroup of $(\mathbb{Q}, +)$
 $(\mathbb{R}, +)$
 $(\mathbb{C}, +)$

$\mathbb{Q}$ subgroup of $(\mathbb{R}, +)$
 $(\mathbb{C}, +)$

$\mathbb{R}$ subgroup of $(\mathbb{C}, +)$

3) $m\mathbb{Z} = \{m \times 1 \mid x \in \mathbb{Z}\}$ subgroup of $(\mathbb{Z}, +)$

4) $H = \{z \in \mathbb{C} \mid |z| = 1\}$ subgroup of $(\mathbb{C}^*, \cdot)$ → circle group which one?
→ not a subgroup of $(\mathbb{C}, +)$

5) $U_n = \{z \in \mathbb{C} \mid z^n = 1\}$ subgroup of $(\mathbb{C}^*, \cdot)$

...

6. SUBRINGS

$(R, +, \cdot)$ ring.

$A \subseteq R$

A subring if: $\begin{cases} 1) A \text{ stable subset} \\ 2) (A, +, \cdot) \text{ ring} \end{cases}$

Theorem

A subring $\Leftrightarrow \begin{cases} A \neq \emptyset \ (0 \in A) \\ (\forall) x, y \in A, \ x - y \in A \\ (\forall) x, y \in A, \ x \cdot y \in A \end{cases}$

7. SUBFIELDS

$(K, +, \cdot)$ field

$A \subseteq K$

A subfield if: $\begin{cases} 1) A \text{ stable subset of } (K, +, \cdot) \\ 2) (A, +, \cdot) \text{ field.} \end{cases}$

Theorem:

A subfield if: $\begin{cases} |A| \geq 2 \ (0, 1 \in A) \\ (\forall) x, y \in A, \ x - y \in A \\ (\forall) x, y \in A, \ y \neq 0, \ x \cdot y^{-1} \in A \end{cases}$

8. HOMOMORPHISM

$(G, \cdot), (G', \cdot)$ groups

$$f: G \rightarrow G'$$

$$f(x \cdot y) = f(x) \cdot f(y), (\forall) x, y \in G$$

9. ISOMORPHISM

→ homomorphism + bijective.

$$G \cong G'$$

↓

$$\text{id. elem} = 1$$

↘

$$\text{id. elem} = 1'$$

Theorem $f: G \rightarrow G'$ group homomorphism

$$1) f(1) = 1'$$

$$2) (f(x))^{-1} = f(x^{-1}), (\forall) x \in G.$$

Examples

10. RING HOMOMORPHISM

$$(R, +, \cdot), (R', +, \cdot), f: R \rightarrow R'$$

$$\begin{cases} f(x+y) = f(x) + f(y) \\ f(x \cdot y) = f(x) \cdot f(y) \end{cases}$$

→ properties: $f(0) = 0'$

$$f(-x) = -f(x)$$

$$f(1) = 1' \quad ?$$

11. RING ISOMORPHISM

→ homomorphism + bijective

$$R \cong R'$$

Examples

12. UNITARY RING HOMOMORPHISM

$$(R, +, \cdot), (R', +, \cdot), f: R \rightarrow R'$$

$$f(1) = 1'$$

Theorem

$(R, +, \cdot), (R', +, \cdot)$ $f: R \rightarrow R'$ ring homom.

$\downarrow$ $\downarrow$
 1 $1'$

1) f surj $\Rightarrow f$ is unitary

2) f is ring isom $\Rightarrow f$ is unitary

3) f is unitary $\left. \begin{array}{l} x \in R \text{ has } x^{-1} \in R \end{array} \right\} \Rightarrow f(x) \text{ has an inv.}$
 $(f(x))^{-1} = f(x^{-1}).$